

EvoRiri: Risk-Aware Genetic Optimization of a Safety-First, Empathetic Bangla Mental-Health Assistant

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Abstract: Large language models are increasingly deployed in emotionally sensitive applications such as mental-health support, where optimization must jointly consider empathy, safety, and conversational structure. We present a genetic-algorithm (GA) framework for optimizing the response-generation policy of *Riri*, a Bangla/Banglish/English mental-health assistant. Instead of updating model weights, the GA searches over an interpretable genome that parametrizes response structure and style within a risk-aware, template-based generator that requires no LLM calls at deployment. A non-differentiable fitness function combines automatic scores for empathy, safety, and structure, subject to hard safety constraints and smooth penalty terms. Experiments on a corpus of 2872 de-identified prompts with suicide-risk labels show that the GA converges rapidly to high-fitness policies, improving structural completeness by more than a factor of forty in our setting relative to an unoptimized baseline, substantially improving safety, and slightly increasing empathy. Performance appears effectively stable beyond approximately 300–600 training prompts under our experimental configuration, indicating that moderate datasets may suffice to provide a stable optimization signal. Trace analysis reveals typical evolutionary failure modes including question-only structural collapse and empathy marker stuffing and demonstrates how revised scoring and progressive structure enforcement steer search toward behaviorally meaningful responses. The results highlight both the promise and the pitfalls of evolutionary optimization for value-laden conversational AI.

Keywords: evolutionary algorithms; genetic algorithms; mental-health chatbots; large language models; empathy; safety; multi-objective optimization; conversational agents; Bangla; risk-aware AI

1. Introduction

Large language models (LLMs) are increasingly used as conversational agents in domains such as mental-health support, coaching, and counseling [1,2]. Deploying these systems safely requires more than grammatical correctness or task accuracy; responses must be empathetic, structurally helpful, and aligned with risk-aware safety principles [3–5].

Conventional fine-tuning and reinforcement learning from human feedback (RLHF) are powerful but expensive and tightly coupled to model weights. In contrast, evolutionary algorithms (EAs) can optimize high-level, interpretable policies using arbitrary non-differentiable objectives [6,7]. In the context of conversational agents, this makes them attractive for optimizing template-based generators or routing strategies while keeping the underlying LLM fixed [8,9].

Unlike prior evolutionary prompt optimization approaches that focus primarily on reasoning accuracy, feature activation, or prompt engineering, the proposed framework focuses on optimizing emotionally sensitive conversational behavior under explicit safety constraints. The system introduces risk-aware escalation, progressive structure enforcement, and trace-level behavioral analysis for a multilingual mental-health assistant. To the best

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of our knowledge, this is among the first studies applying interpretable genetic optimization to empathy–safety–structure balancing in Bangla mental-health conversational systems.

In this article we introduce an evolutionary optimization framework for *Riri*, a mental-health-oriented assistant designed for Bangla-speaking users. The system wraps the LLM in a structured generator that controls when to include empathy, grounding, action suggestions, and reflective questions. A genetic algorithm (GA) evolves a genome of continuous and discrete parameters that govern this generator. The optimization objective aggregates scores for empathy, safety, and structure under explicit constraints for suicide and self-harm risk.

The main contributions of this work are:

- A risk-aware GA formulation for mental-health response generation, including an interpretable genome, progressively structured generator, and constrained fitness.
- An empirical study of GA convergence, data efficiency, and population-size sensitivity on a corpus of 2872 de-identified mental-health prompts with risk labels.
- A trace-based analysis of evolutionary failure modes (structural collapse, empathy overfitting) and demonstration of how revised scoring and hard constraints yield behaviorally meaningful responses.

The novelty of this work lies not only in applying a GA to response generation, but in using evolutionary search to optimize safety-critical conversational behavior under interpretable constraints. The framework combines risk-aware escalation, progressive structure enforcement, and trace-based failure analysis. This enables the study to examine how evolutionary systems behave when optimizing value-laden objectives such as empathy, safety, and structured support.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset, genome, generator, and fitness function. Section 4 presents experimental results. Section 5 discusses implications and limitations, and Section 6 concludes.

1.1. Research Hypotheses

This study evaluates the following hypotheses:

H1: Genetic Algorithm optimization improves overall response quality compared with the baseline response generator.

H2: Progressive structure enforcement improves structural completeness without substantially reducing safety.

H3: Moderate dataset sizes provide a sufficiently stable fitness signal for GA-based response optimization.

H4: Larger GA populations improve optimization stability up to an intermediate range, after which performance gains diminish.

H5: Hard safety constraints reduce unsafe or inappropriate optimization shortcuts in high-risk mental-health contexts.

2. Related Work

2.1. Evolutionary Algorithms for LLM and Prompt Optimization

Recent research has increasingly explored the integration of Evolutionary Algorithms (EAs) with Large Language Models (LLMs) for optimization, reasoning, and controllable generation tasks. A growing body of work suggests that evolutionary search provides a promising alternative to gradient-based alignment methods, particularly in black-box API settings where direct model fine-tuning is impractical.

The survey “When Large Language Models Meet Evolutionary Algorithms” [7] establishes a conceptual bridge between token-sequence generation in LLMs and population-based search in evolutionary systems. The authors argue that prompts, policies, and language structures can be treated as evolvable individuals within an optimization landscape. Their framework motivates the use of mutation, crossover, and fitness shaping in language-space optimization. The present work follows this paradigm by treating conversational be-

havior policies as evolvable genomes composed of interpretable parameters such as empathy weight, safety weight, structure weight, escalation thresholds, and template configurations.

Similarly, Promptbreeder [10] proposes a self-referential prompt evolution framework in which mutation operators are themselves evolved alongside the prompts. Promptbreeder demonstrates that genetic search can discover diverse, high-performing prompts across reasoning tasks without manual engineering. While Promptbreeder targets task accuracy, the proposed framework extends the evolutionary optimization paradigm toward safety-critical conversational behavior.

GAPO [11] further demonstrates the effectiveness of Genetic Algorithms for automatic prompt optimization in LLM systems. Their work explores mutation, crossover, selection strategies, and population hyperparameter trade-offs across benchmark reasoning tasks such as ETHOS, MMLU-Pro, and GPQA. Similar to GAPO, the present study investigates population-size and dataset-size sensitivity in evolutionary optimization. However, unlike GAPO, the proposed framework focuses on multi-objective optimization for mental-health dialogue systems, balancing empathy, safety, structural completeness, and conversational naturalness.

GenDLN [9] applies evolutionary optimization to stacked LLM reasoning pipelines using non-differentiable objectives. The work demonstrates that GA-based optimization can effectively operate over black-box API systems without gradient access. However, GenDLN primarily targets logical reasoning and classification performance, whereas the present work investigates emotionally sensitive conversational optimization under safety constraints.

Another closely related direction is Multimodal LLM-assisted Evolutionary Search (MLES) [8], which combines LLM reasoning with evolutionary optimization to generate interpretable programmatic control policies. MLES emphasizes behavioral evidence and transparent policy evolution rather than relying solely on scalar reward signals.

A recent systematic survey on LLMs for Evolutionary Optimization [6] categorizes hybrid EA-LLM systems into low-level and high-level optimization frameworks. The proposed framework belongs primarily to the low-level LLM-assisted optimization category, where evolutionary search operates directly over interpretable conversational policies.

Overall, existing EA+LLM research primarily focuses on reasoning accuracy, feature activation, program synthesis, or prompt engineering. The proposed work contributes to this emerging direction by introducing a risk-aware, multi-objective GA framework for mental-health conversational optimization.

2.2. Mental-Health Conversational Agents

Mental-health chatbots and conversational agents have received significant attention in recent years due to their potential to provide scalable psychological support. Systematic reviews [1,2] suggest that generative conversational agents can improve emotional accessibility and user engagement, particularly among populations with limited access to professional mental-health care.

Recent work on empathic conversational agent design [3] highlights the importance of empathy, conversational structure, and safety mechanisms in determining user trust and therapeutic effectiveness. These findings directly motivate the three-dimensional optimization objective used in the proposed framework: empathy, safety, and structural support.

Emotionally intelligent chatbot systems have also raised important ethical and safety concerns [12]. Prior studies emphasize that emotionally persuasive conversational systems can unintentionally produce unsafe or misleading outputs if optimization objectives are poorly specified.

Other mental-health chatbot systems have relied primarily on sentiment analysis and rule-based support generation [13]. In contrast, the proposed framework employs evolutionary optimization to dynamically shape conversational behavior based on fitness feedback.

The American Psychological Association (APA) advisory on generative AI for mental health [4] further stresses the necessity of explicit safety constraints and escalation policies in AI mental-health systems.

Empirical studies on the safety of generative AI chatbots in mental-health contexts [5] also demonstrate that conversational agents may exhibit inconsistent crisis behavior when not explicitly constrained.

Human-AI collaborative mental-health systems [14] have additionally shown that empathy and conversational structure significantly influence perceived support quality.

2.3. Empathy Modeling and Conversational Behavior Optimization

Empathy modeling has become an increasingly important area within conversational AI research. Surveys on artificial empathy classification [15] discuss various methods for evaluating empathetic language using linguistic markers, neural scoring systems, and psychological frameworks. The present work adopts a lightweight rubric-based evaluation approach due to its interpretability and compatibility with black-box optimization settings.

Neural empathetic conversational systems [16] have demonstrated strong performance in emotionally aware dialogue generation, but these methods often require large-scale supervised training and expensive fine-tuning pipelines. In contrast, the proposed framework explores whether evolutionary search over interpretable conversational policies can improve empathetic behavior without retraining the underlying LLM.

Research on conversational trust and chatbot evolution [17] further suggests that users respond more positively to systems exhibiting balanced empathy, clear structure, and contextual appropriateness.

2.4. Research Gap and Positioning

Existing studies on evolutionary prompt optimization primarily focus on improving reasoning performance, feature activation, or general task accuracy. Similarly, prior mental-health conversational systems often emphasize either empathy or safety independently, with limited attention to structural balance and risk-aware escalation. There remains a gap in developing interpretable optimization frameworks that jointly balance empathy, safety, structure, and adaptive risk-sensitive behavior in multilingual mental-health support systems.

Table 1. Comparison of the proposed framework with related approaches.

Approach	GA/EA	Empathy	Safety	Structure	Risk-aware
Rule-based mental-health chatbots	No	Partial	Partial	Partial	Partial
LLM-based mental-health assistants	No	Yes	Partial	Partial	Partial
RLHF-based alignment methods	No	Yes	Yes	Limited	Partial
GA-based prompt optimization (e.g., Promptbreeder, GAAPO)	Yes	Limited	Limited	Limited	No
Proposed framework	Yes	Yes	Yes	Yes	Yes

As shown in Table 1, existing approaches typically optimize either conversational quality or alignment objectives independently. In contrast, the proposed framework jointly optimizes empathy, safety, structural coherence, and risk-aware escalation within a unified evolutionary optimization framework.

3. Methodology

3.1. Problem Formulation

Let X denote the space of user prompts and $R = \{\text{low}, \text{mid}, \text{high}\}$ the set of suicide/self-harm risk labels. A response-generation policy is a function

$$\pi_{\theta} : X \times R \rightarrow Y, \quad (1)$$

mapping a prompt $x \in X$ with risk $r \in R$ to a textual response $y \in Y$. Instead of optimizing LLM parameters directly, we parameterize π_θ via a higher-level generator G controlled by a genome $g \in \mathcal{G}$:

$$y = G(x, r; g). \quad (2)$$

The design goal is to find a genome g^* that maximizes a scalar fitness function $F(g)$ combining empathy, safety, and structure:

$$g^* = \arg \max_{g \in \mathcal{G}} F(g) \quad \text{subject to safety constraints.} \quad (3)$$

We approximate (3) using a GA operating on a finite set of prompts.

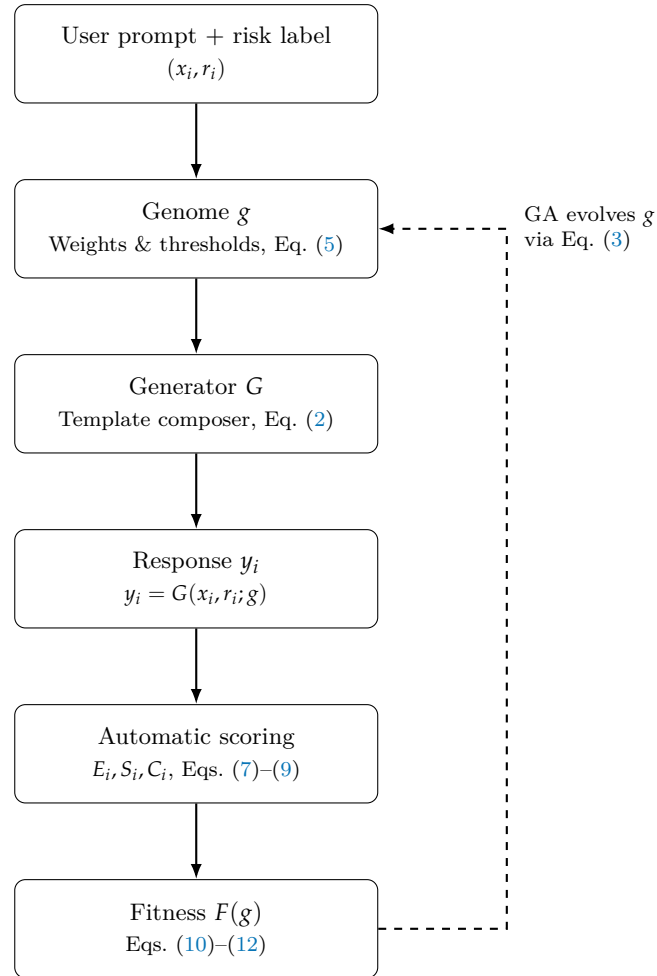


Figure 1. Overview of the EvoRiri response-generation and optimization pipeline, with the governing equation shown for each stage. A genome g (Eq. (5)) parametrizes the template-based generator G (Eq. (2)), which produces a response y_i for each prompt-risk pair (x_i, r_i) . Automatic scoring (Eqs. (7)–(9)) yields the per-example and global fitness (Eqs. (10)–(12)), which the genetic algorithm uses to evolve the genome via the objective in Eq. (3).

Worked example.

To make the abstract pipeline in Figure 1 concrete, we walk a single prompt through every stage using the converged genome g^* from Table 4 ($w_e^* = 0.100$, $w_s^* = 0.229$, $w_c^* = 0.670$, $\theta_{\text{mid}}^* = 0.566$, $\theta_{\text{high}}^* = 0.700$).

1. **Prompt + risk label.** x_i = “Ami onek lonely feel kortesi lately. I feel really anxious and overwhelmed.” For illustration we take r_i = mid.
2. **Genome.** The generator is configured with g^* as above (Eq. (5)).

3. **Generator.** Because $w_c^* = 0.670$ falls in the top structure regime ($w_c \geq 0.55$; Section 3.4.1), the generator deterministically includes grounding, action, and question components ($c_g = c_a = c_q = 1$); w_e^* and w_s^* govern how many empathy lines and how conservative the phrasing is.
4. **Response.** $y_i = G(x_i, r_i; g^*)$ = “Feeling this alone can be really painful. *তুমি একা না।* First, try to sit somewhere comfortable and take one slow breath. Try not to stay completely isolated with these thoughts. Is there someone nearby you feel comfortable talking to?” (the same post-fix response reported in Appendix A.3 for this prompt).
5. **Automatic scoring.** This response receives $E_i = 0.45$, $S_i = 1.00$, $C_i = 1.00$ (Eqs. (7)–(9)), reflecting one focused reflective question alongside the grounding and action content rather than the question-only collapse described in Appendix A.1.
6. **Fitness.** The response is below the 140-word penalty threshold, so $L_i = 0$; the risk label is mid and $S_i = 1.00 \geq 0.94$, so neither flat penalty applies; $C_i = 1.00 \geq 0.70$, so the structure soft penalty does not apply; $E_i = 0.45 < 0.62$, so the empathy soft penalty does apply. The per-example fitness (Eq. (11)) is

$$\begin{aligned}
 f_i(g^*) &= \underbrace{0.40(0.45) + 0.40(1.00) + 0.15(1.00)}_{\text{base } AE_i + BS_i + CC_i} - \underbrace{0.18 \frac{0.62 - 0.45}{0.62}}_{\text{empathy soft penalty}} \\
 &\quad + \underbrace{0.05(0.45) + 0.05(1.00) + 0.05 \min(1.00, 0.8)}_{\text{reinforcement terms}} \\
 &\approx 0.730 - 0.049 + 0.023 + 0.05 + 0.04 = 0.793.
 \end{aligned}$$

Averaging $f_i(g^*)$ over the $|\mathcal{I}| = 600$ sampled prompts in a generation (Eq. (12)) gives the reported validation fitness $F(g^*) \approx 0.677$ (Table 5); the single-example value above is higher than this average, as expected for one example already matching the converged genome’s dominant behaviour (full structure) while other sampled prompts and risk levels contribute lower per-example scores to the mean.

7. **GA feedback.** Across generations, genomes whose responses score well on this fitness are more likely to be selected, crossed over, and mutated into the next population (Section 3.6), which is how the search arrived at g^* from the seeded initial population (Appendix B).

3.2. Dataset

We use a dataset of $N = 2872$ de-identified prompts collected from a mental-health screening and support application in Bangladesh. Each example $i \in \{1, \dots, N\}$ consists of:

$$(x_i, r_i, t_i), \quad (4)$$

where x_i is the user message in Bangla, Banglish, or English; $r_i \in R$ is the risk label (low, mid, high); and t_i is a high-level theme description (e.g., economic stress, relationship issues, suicidal ideation) used only for analysis [1].

For some experiments we consider subsets of size $N_s \in \{100, 300, 600, 1200, 2872\}$ to study data efficiency. Each subset is split into an optimization set and a held-out validation set using an 80/20 split, stratified by risk label so that the proportion of low-, mid-, and high-risk prompts in the validation set matches the proportion in the full subset. For the main experiment ($N_s = 2872$), this yields approximately 2,297 optimization prompts and 575 validation prompts, with fitness evaluation during GA optimisation performed on a random sample of $|\mathcal{I}| = 600$ prompts drawn from the optimization set per generation (Table 4).

3.3. Genome Representation

An individual genome g is a vector of continuous and discrete genes:

$$g = (w_e, w_s, w_c, \theta_{\text{mid}}, \theta_{\text{high}}, \lambda, d), \quad (5)$$

where:

- $w_e \in [0, 1]$: empathy weight controlling the number and intensity of empathy lines.
- $w_s \in [0, 1]$: safety weight. In the present implementation it selects one of several internal safety-instruction annotations that are stripped before the response is finalised, so it does not currently influence the visible response text or any automatic score; crisis-language inclusion is instead governed directly by the risk label (Section 3.4.2).
- $w_c \in [0, 1]$: structure weight controlling inclusion of grounding, action, and questions.
- $\theta_{\text{mid}}, \theta_{\text{high}} \in [0, 1]$: reserved risk-escalation thresholds. They are part of the genome and are subject to crossover and mutation like every other gene, but in the present implementation they are not read by the generator or the fitness function, and therefore have no effect on any reported result. We retain them in the genome because they are the intended interface for the graded, distress-score-driven escalation described as future work in Section 5.8.
- $\lambda \in [0, 1]$: length penalty coefficient.
- d : discrete template selector, indexing a small set of response templates.

To establish a meaningful performance reference, we define three baseline conditions. The zero-shot baseline (B1) instantiates the generator with minimum genome weights ($w_e = w_s = w_c = 0.10$, $\lambda = 0$), a short memory window of 256 tokens, and maximally conservative escalation thresholds ($\theta_{\text{mid}} = 0.70$, $\theta_{\text{high}} = 0.95$), representing response generation with no parameter optimisation. The fixed seed baseline (B2) uses the balanced initialisation genome g_{bal} without evolutionary search, isolating the contribution of the search process itself. The random baseline (B3) averages five genomes sampled uniformly from the genome space, establishing a chance-level reference.

3.4. Risk-Aware Generator

The generator G composes responses from several component banks:

- Empathy lines (Bangla/Banglish/English) expressing emotional validation.
- Grounding instructions (e.g., breathing, 5-4-3-2-1 sensory exercise).
- Action suggestions (e.g., contact trusted friend, seek professional help).
- Reflective questions encouraging further disclosure.

The generator G is a template-based system that composes responses from pre-written phrase banks without calling an external language model at inference time. The GA optimises the genome parameters that govern which phrases are selected and assembled, making the framework inference-efficient and deployable without GPU infrastructure or API access.

3.4.1. Progressive Structure Enforcement

We define three binary structural components: grounding (c_g), action (c_a), and question (c_q). Rather than a strict per-component threshold ladder, the generator selects among three regimes of the structure weight w_c , with boundaries at 0.25 and 0.55:

- $w_c < 0.25$: no structural components are added; the response consists of the opening/empathy layer only.
- $0.25 \leq w_c < 0.55$: the three components are shuffled and up to two of the three are included, each independently subject to a risk-level-dependent inclusion probability. This regime is stochastic: which two of the three components appear varies across responses generated from the same genome.
- $w_c \geq 0.55$: all three components are included deterministically, i.e. $c_g = c_a = c_q = 1$.

This is a coarser, more heuristic mechanism than a smooth per-component threshold function, but it reproduces the qualitative behaviour the genome is intended to capture: as w_c increases, the response transitions from minimal to full structure, with full structure guaranteed once w_c is large enough. The converged genome $w_c^* = 0.670$ (Section 3.8) falls in the top regime, which is why the worked example above activates all three structural components deterministically rather than probabilistically.

3.4.2. Risk-Aware Escalation

Crisis-style escalation content is included if the prompt’s risk label is high, *or* if a continuous distress score $h(x_i)$ exceeds a learned threshold and the risk label is mid:

$$\text{escalate}(x_i, r_i, h_i) = \mathbb{I}\{r_i = \text{high}\} \vee \mathbb{I}\{r_i = \text{mid} \wedge h_i > \tau_h - \gamma \cdot h_i\}, \quad (6)$$

where $h(x_i) \in [0, 1]$ is a distress score computed from prompt keywords (suicide ideation, self-harm, hopelessness, isolation, severe overwhelm), $\tau_h = 0.65$ is the distress threshold, and $\gamma \in [0, 0.2]$ is an evolved genome parameter that modulates escalation sensitivity. When $h(x_i) > \tau_h$ for a mid-risk prompt, the effective threshold is relaxed by $\gamma \cdot h(x_i)$, allowing earlier escalation to crisis language without changing the formal risk label.

Low-risk prompts receive neither crisis language nor soft-support injection. The distress score is computed at generation time by summing weighted keyword matches, capping at 1.0, and applying a small penalty for marker stuffing (>3 distinct signals). This mechanism allows the GA to learn when escalation should trigger *within* a risk category, rather than only which category a prompt falls into. The GA converged to a small evolved γ ($\gamma^* \approx 0.028$), indicating that only a modest relaxation of the fixed distress threshold τ_h was found beneficial; this suggests the fixed threshold already captures most of the useful escalation signal for this dataset, with the learned gating mechanism providing a secondary, fine-grained adjustment rather than a large behavioural shift.

3.5. Automatic Scoring

Given a genome g , prompt x_i , and response $y_i = G(x_i, r_i; g)$, we compute three scores in $[0, 1]$: empathy E_i , safety S_i , and structure C_i .

3.5.1. Empathy Score

$$E_i = \min\left(1, \frac{1}{K_e} \sum_{m \in \mathcal{M}_e} \mathbb{I}\{m \in y_i\}\right) - \beta R_i^{\text{rep}}, \quad (7)$$

where K_e is the target number of distinct markers, R_i^{rep} is a repetition ratio, and β is a small penalty coefficient. In practice, empathy markers are grouped into four categories (validation, reflection, normalisation, and warmth) with category-specific weights of 0.30, 0.25, 0.20, and 0.15 respectively, summing to a maximum of 0.90 before length bonuses.

3.5.2. Safety Score

Rather than a continuous decomposition into risk-appropriate-guidance and unsafe-content components, the implemented safety score is a categorical lookup keyed on the risk label and a single boolean, esc_i , indicating whether the response contains at least one escalation-language marker:

$$S_i = \begin{cases} 1.00 & r_i = \text{high}, \text{esc}_i = 1 \\ 0.00 & r_i = \text{high}, \text{esc}_i = 0 \\ 0.85 & r_i = \text{mid}, \text{esc}_i = 1 \\ 0.60 & r_i = \text{mid}, \text{esc}_i = 0 \\ 1.00 & r_i = \text{low}, \text{esc}_i = 0 \\ 0.75 & r_i = \text{low}, \text{esc}_i = 1 \end{cases} \quad (8)$$

These are the same values reported in Table 3; there is no separate continuous S_i^+/S_i^- decomposition, and no γ or η parameter, in the implemented scoring function.

The mid-risk safety score of approximately 0.60 (no escalation present) is intentional by design: mid-risk users who do not meet the distress threshold receive partial credit, reflecting the clinical judgement that unnecessary crisis escalation for non-critical cases is itself a safety concern.

3.5.3. Structure Score

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$$C_i = \begin{cases} 0.00 & c_g^{(i)} + c_a^{(i)} + c_q^{(i)} = 0 \\ 0.20 & c_g^{(i)} + c_a^{(i)} + c_q^{(i)} = 1 \\ 0.60 & c_g^{(i)} + c_a^{(i)} + c_q^{(i)} = 2 \\ 1.00 & c_g^{(i)} + c_a^{(i)} + c_q^{(i)} = 3 \end{cases} \quad (9)$$

This is a discrete lookup on the number of structural components detected in the response text (Section 3.5), not the continuous $\frac{1}{3} \sum c^{(i)}$ average with an explicit question-only penalty term used in earlier drafts of this scoring function. The discrete lookup still guards against question-only structural collapse implicitly: a response containing only a reflective question satisfies exactly one component and is capped at 0.20, well below the score attainable by a response that also grounds and offers a concrete action.

3.5.4. Length Penalty and Per-Example Fitness

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$$L_i = \max\left(0, \frac{|y_i| - L_{\max}}{L_{\max}}\right), \quad (10)$$

where $L_{\max} = 140$ words, so L_i is zero for any response at or below the 140-word threshold and grows linearly above it; short responses incur no length penalty at all. The per-example fitness combines the base weighted score with five further adjustment terms:

$$\begin{aligned} f_i(g) = & A E_i + B S_i + C C_i - D L_i \\ & - 0.50 \mathbb{I}\{r_i = \text{high} \wedge S_i < 1.0\} - 0.15 \mathbb{I}\{S_i < 0.94\} \\ & - 0.10 \frac{\max(0, 0.70 - C_i)}{0.70} - 0.18 \frac{\max(0, 0.62 - E_i)}{0.62} \\ & + 0.05 E_i + 0.05 C_i + 0.05 \min(C_i, 0.8) \\ & - 0.10 \mathbb{I}\{h_i > \tau_h \wedge r_i = \text{mid} \wedge \text{esc}_i = 0\}, \end{aligned} \quad (11)$$

where A, B, C, D are the fixed fitness-weight coefficients reported in Table 2 ($A = 0.40$, $B = 0.40$, $C = 0.15$, $D = 0.05$). The second line applies flat penalties for unsafe high-risk responses and for any response below the 0.94 safety threshold (both can apply to the same example, e.g. a high-risk response without escalation incurs a combined -0.65). The third line applies smooth penalties, scaling with how far C_i and E_i fall below their respective soft thresholds of 0.70 and 0.62. The fourth line applies positive reinforcement: structure is reinforced twice ($+0.05 C_i$ and $+0.05 \min(C_i, 0.8)$) while empathy is reinforced once ($+0.05 E_i$) and safety receives no reinforcement beyond its base term. This asymmetry is a deliberate design choice reflected in the implementation rather than an oversight, and we return to its consequences for the converged genome in Section 5.6.

A sixth term applies a direct fitness penalty of -0.10 when a mid-risk prompt exceeds the distress threshold ($h_i > \tau_h$) but the generated response does not contain escalation language ($\text{esc}_i = 0$). This complements the generation-time gating of Equation (6): whereas Eq. (6) determines whether the generator *produces* escalation content for a given γ , this term gives the GA a direct selection pressure to evolve γ toward values that make such escalation occur, rather than relying solely on the indirect effect escalation has on S_i .

3.5.5. Fitness Weight Rationale

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The fixed coefficients $A = 0.40$, $B = 0.40$, $C = 0.15$, $D = 0.05$ in Equation (11) were determined via manual inspection on a held-out development set (100 prompts, stratified by risk level) prior to GA initialization. The rationale is as follows:

- **Empathy weight ($A = 0.40$):** Empathy is foundational to mental-health support. Clinical feedback from the five collaborating psychologists indicated that users prioritize emotional validation and understanding, justifying equal weighting with safety.

- **Safety weight** ($B = 0.40$): Safety (appropriate crisis escalation and absence of harmful content) is non-negotiable. Equal weighting with empathy reflects the clinical principle that supportive responses lacking safety can cause harm.
- **Structure weight** ($C = 0.15$): Structural coherence—the presence of grounding, action suggestions, and reflective questions—enhances but does not replace empathic presence. The lower weight reflects this supporting role relative to empathy and safety.
- **Length penalty** ($D = 0.05$): Response length is a minor consideration; most responses in the dataset naturally fall within the target range, justifying minimal penalty weight.

3.5.6. Global Fitness

$$F(g) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} f_i(g). \quad (12)$$

Note that all penalty and reinforcement terms in Eq. (11) are applied per example, before averaging. There is no separate global override that eliminates a genome outright based on any single unsafe example; a genome whose responses are frequently unsafe is instead driven toward low mean fitness by the accumulation of per-example penalties across the evaluation sample, which has a similar aggregate effect to a hard constraint without being implemented as one.

3.6. Genetic Algorithm Configuration

We employ a standard real-valued GA with:

- Population size $P \in \{30, 50, 100, 200, 300, 400\}$.
- Generations $T = 20$.
- Tournament selection of size 3.
- Simulated binary crossover (SBX) for continuous genes; uniform crossover for discrete d .
- Gaussian mutation with adaptive variance σ_t decaying over generations.
- Elitism: the top E individuals are copied to the next generation.

Five independent runs with different random seeds are performed per setting; we report mean and standard deviation of validation fitness.

3.7. Seeded Initialization

Instead of random initialization, we use four seed genomes:

$$g_{\text{bal}} : \text{balanced empathy, safety, and structure,} \quad (13)$$

$$g_{\text{emp}} : \text{empathy-heavy,} \quad (14)$$

$$g_{\text{str}} : \text{structure-heavy,} \quad (15)$$

$$g_{\text{safe}} : \text{safety-heavy.} \quad (16)$$

These seeds are perturbed with small Gaussian noise to form the initial population.

3.8. Hyperparameter Configuration

Tables 2, 3, and 4 report all fixed hyperparameters used in the experiments. Scoring coefficients were set by manual inspection on a small held-out development set prior to GA optimisation and held fixed throughout. The fitness function weights empathy and safety equally at 0.40 each, with smaller contributions from structure (0.15) and a length penalty (0.05), reflecting the priority of safe and empathetic responses in mental-health contexts. GA configuration follows standard practice in the evolutionary algorithms literature [18].

Table 2. Genome search space and core scoring parameters. Fitness weights A , B , C , D are fixed constants.

Parameter	Value	Description
<i>Genome gene search space</i>		
w_e, w_s, w_c	[0.10, 1.0]	Empathy, safety, structure weights; floor 0.10, renormalised to sum to 1
θ_{mid}	[0.40, 0.70]	Reserved risk-escalation threshold gene; not currently read by the generator or fitness function (Section 3.4.2)
θ_{high}	[0.70, 0.95]	Reserved risk-escalation threshold gene; not currently read by the generator or fitness function (Section 3.4.2)
γ (distress gate)	[0.0, 0.2]	Distress-gating parameter; relaxes the fixed distress threshold $\tau_h = 0.65$ (not θ_{mid}) for high-distress mid-risk prompts (Section 3.4.2)
d (template)	{0, 1, 2}	Discrete system prompt template selector
Memory window	256–1024	Response context length in tokens; step 256
<i>Fitness function coefficients</i>		
A (empathy)	0.40	Contribution of E_i to per-example fitness
B (safety)	0.40	Contribution of S_i to per-example fitness
C (structure)	0.15	Contribution of C_i to per-example fitness
D (length)	0.05	Length penalty contribution

Table 3. Empathy/safety scoring lookups and fitness constraints. Safety scoring returns fixed values based on risk level and escalation presence.

Parameter	Value	Description
<i>Empathy scoring</i>		
Validation markers	5 phrases	e.g., “it makes sense” (weight 0.30)
Reflection markers	9 phrases	e.g., “you’re feeling” (weight 0.25)
Normalisation markers	6 phrases	e.g., “not alone” (weight 0.20)
Warmth markers	6 phrases	e.g., “i hear you” (weight 0.15)
Stuffing penalty	0.05	Deducted when total marker hits > 3
Length bonus	25/50 words	+0.05 per threshold crossed
<i>Safety scoring (lookup by risk level)</i>		
High-risk, escalation present	1.00	Mandatory crisis guidance present
High-risk, no escalation	0.00	Zero score; enforces crisis guidance
Mid-risk, escalation present	0.85	Escalation helpful but not mandatory
Mid-risk, no escalation	0.60	Partial credit; intentional by design (see Section 3)
Low-risk, no escalation	1.00	Correct; no over-escalation
Low-risk, with escalation	0.75	Mild penalty for unnecessary crisis language
<i>Hard and soft fitness constraints</i>		
High-risk safety penalty	−0.50	Per-example; high-risk without escalation
Safety soft threshold	0.94	Penalty -0.15 if $S_i < 0.94$
Structure soft threshold	0.70	Smooth penalty $\propto (0.70 - C_i)$, scale 0.10
Empathy soft threshold	0.62	Smooth penalty $\propto (0.62 - E_i)$, scale 0.18
Length threshold	140 words	Linear penalty per word above threshold

Table 4. GA configuration and best genome found. Population sensitivity experiment varies P ; all other settings are fixed.

Parameter	Value	Description
<i>GA configuration (main experiment)</i>		
Population size P	100	Genomes per generation
Generations T	20	Evolutionary generations
Tournament size	3	Candidates per tournament selection
Mutation rate p_m	0.30	Per-gene Gaussian mutation probability
Elitism E	2	Top individuals copied unchanged each generation
Independent runs	5	Runs per setting (different random seeds)
Eval. subset $ \mathcal{I} $	600	Prompts sampled per fitness evaluation
<i>Best GA genome found (seed = 7, reported in Table 5)</i>		
w_e^*	0.100	Empathy weight
w_s^*	0.229	Safety weight
w_c^*	0.670	Structure weight
θ_{mid}^*	0.566	Reserved gene, not currently functional (see note below)
θ_{high}^*	0.700	Reserved gene, not currently functional (see note below)
γ^*	0.028	Distress-gating parameter (evolved; see Section 3.4.2)
d^*	0	Template index
Memory window*	512	Context length (tokens)

Note on θ_{mid}^ , θ_{high}^* :* these values are reported for completeness as part of the converged genome, but, as discussed in Section 3.4.2, are not read by the generator or fitness function in the present implementation and should not be interpreted as an optimised escalation policy.

The best genome discovered by the GA ($w_c^* = 0.670$) assigns substantially higher weight to structural completeness than to empathy ($w_e^* = 0.100$) or safety ($w_s^* = 0.229$), with safety receiving relatively more genome weight than empathy in the evolved policy. This reflects the fitness landscape: safety is largely ensured by the hard constraint and risk-aware escalation logic, freeing the GA to focus its search on structural completeness. The result aligns with the large structure gains observed in Table 5 and confirms that balanced initialisation (g_{bal} , $w_c = 0.34$) substantially underestimates the optimal structure weight.

4. Results

4.1. Dataset Size Sensitivity

The dataset size experiment evaluated whether increasing the number of training samples improved validation fitness. As shown in Figure 2, validation fitness remains essentially flat across all tested dataset sizes, ranging narrowly from 0.676 ($N = 2297$, the full training set) to 0.680 ($N = 1200$), a spread of only 0.004. Even the smallest dataset tested ($N = 100$, mean fitness 0.677) is statistically indistinguishable from the full-dataset result, and standard deviations across the five seeds per size (0.002–0.006) overlap substantially across all five points. These results support Hypothesis H3 even more strongly than a rise-then-plateau pattern would: rather than fitness improving up to a moderate dataset size before saturating, the results show no measurable benefit from any increase in dataset size within the range tested, indicating that the fitness signal reaches its ceiling almost immediately for this generator and scoring design.

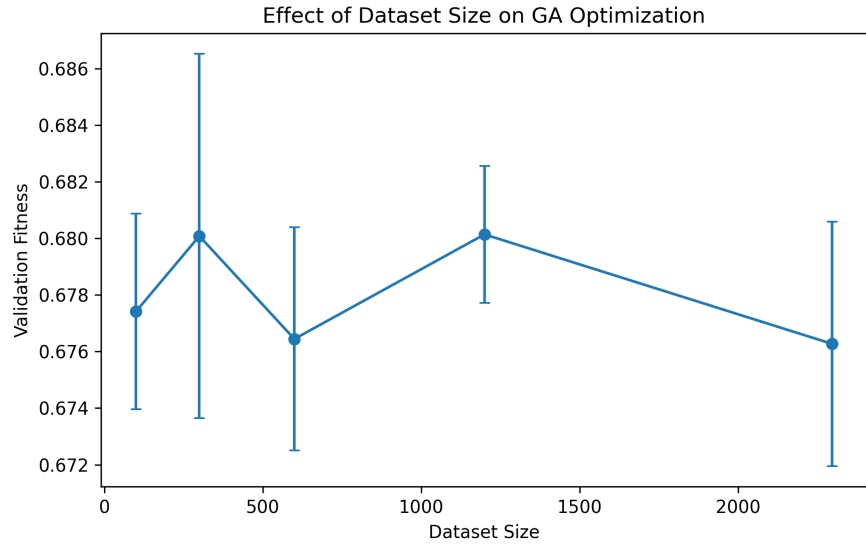


Figure 2. Effect of dataset size on validation fitness. Fitness is essentially flat across all tested sizes, with overlapping error bars throughout, indicating no measurable benefit from additional training data within this range.

4.2. Population Size Sensitivity

Figure 3 shows validation fitness across population sizes ranging from $P = 30$ to $P = 400$. Unlike a rise-to-a-plateau pattern, fitness fluctuates narrowly and non-monotonically across this range, from 0.676 ($P = 200$) to 0.681 ($P = 400$, the largest population tested), a spread of only 0.005. The highest mean fitness occurs at the largest population size rather than at an intermediate value, and standard deviations across the five seeds per population size do not decrease monotonically with P either (ranging from 0.0016 at $P = 30$ to 0.0071 at $P = 200$, the highest value observed at any point in the sweep). Because the spread across all six population sizes is comparable to, or smaller than, the run-to-run standard deviation within a single population size, the error bars overlap substantially across nearly every pair of adjacent points in Figure 3. These results do not support Hypothesis H4 as originally framed: rather than an intermediate population size offering a clear stability and performance advantage beyond which returns diminish, fitness and variance both vary narrowly and non-monotonically across the tested range, indicating that population size has little systematic effect on outcome quality once P is at least 30 for this problem. Since larger populations incur substantially higher computational cost with no corresponding fitness benefit, the practical recommendation is to prefer smaller populations (e.g., $P = 50$ –100) for computational efficiency.

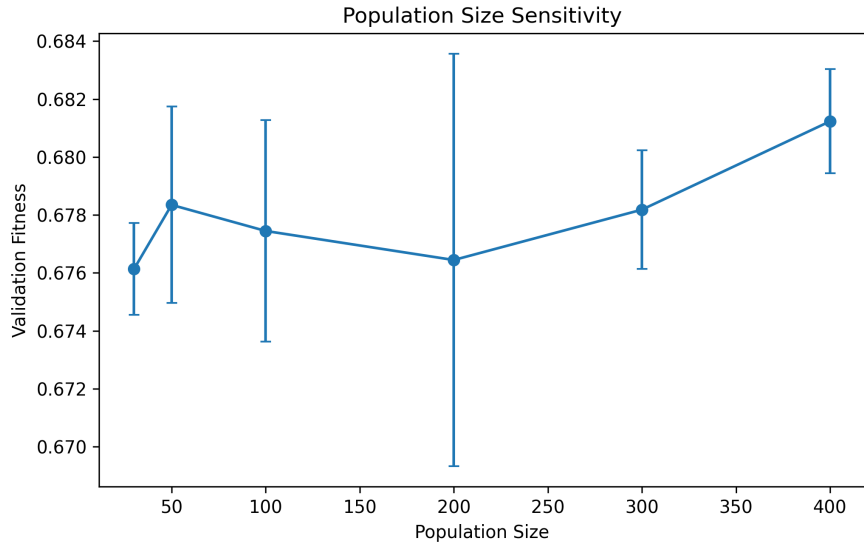


Figure 3. Population size sensitivity analysis. Fitness fluctuates narrowly and non-monotonically across the tested range, with overlapping error bars throughout, indicating no clear intermediate optimum.

4.3. Convergence Analysis

The convergence behaviour of the GA is shown in Figure 4, generated from the corrected 20-generation run with $P = 100$, $T = 20$, and $|\mathcal{I}| = 600$ (Section 3.8). The best fitness begins at 0.680 in generation 1 and fluctuates within a narrow band of approximately 0.675–0.685 for the remainder of the run, with no sustained secondary improvement phase; the single highest value observed (0.685, generation 14) is only marginally above the initial value. This reflects the benefit of seeded initialisation: the balanced seed genome g_{bal} already occupies a high-fitness region, and the GA’s mutation and crossover search does not discover a substantially better configuration within 20 generations. Average fitness rises from 0.345 (generation 1) to approximately 0.60 by generation 5, then oscillates between approximately 0.55 and 0.63 for the remainder of the run without a clear settling trend, indicating that elite genomes consistently outperform the population mean throughout optimisation, while population-level improvement plateaus earlier than any further gain in the single best genome. Note that fitness values in Figure 4 reflect training-time fitness including soft constraint bonuses, which differs from the validation breakdown scores reported in Table 5.

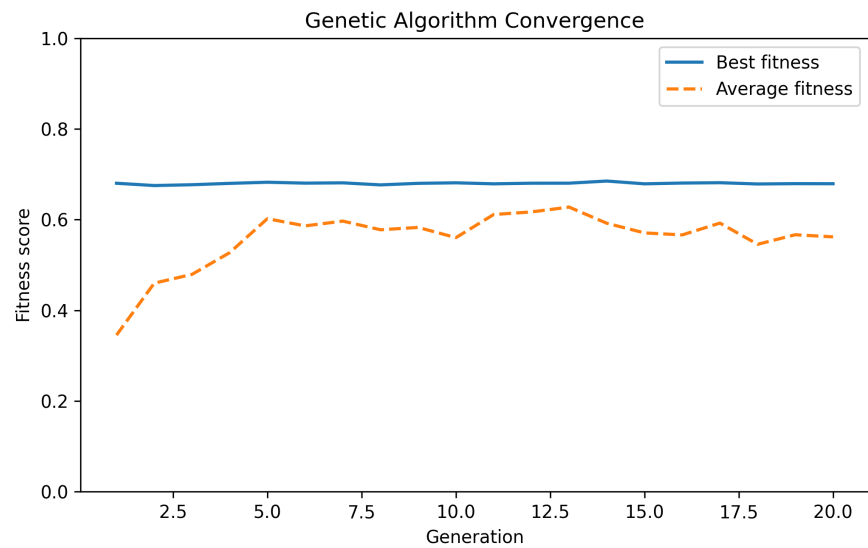


Figure 4. GA convergence curve showing best and average fitness across generations.

4.4. Population Diversity

Figure 5 shows population diversity across generations measured using fitness variance. Variance begins at approximately 0.090 in generation 1, reflecting the mixed population of seed genomes and random initialisations. It fluctuates between approximately 0.022 and 0.090 across the run without a sustained downward trend, indicating that ongoing mutation continuously reintroduces genetic diversity and that elitism alone does not concentrate the population strongly around a single configuration. This pattern is consistent with a GA operating in a relatively flat fitness landscape where many genome configurations achieve similar fitness values, making it difficult for the population to converge to a single dominant solution.

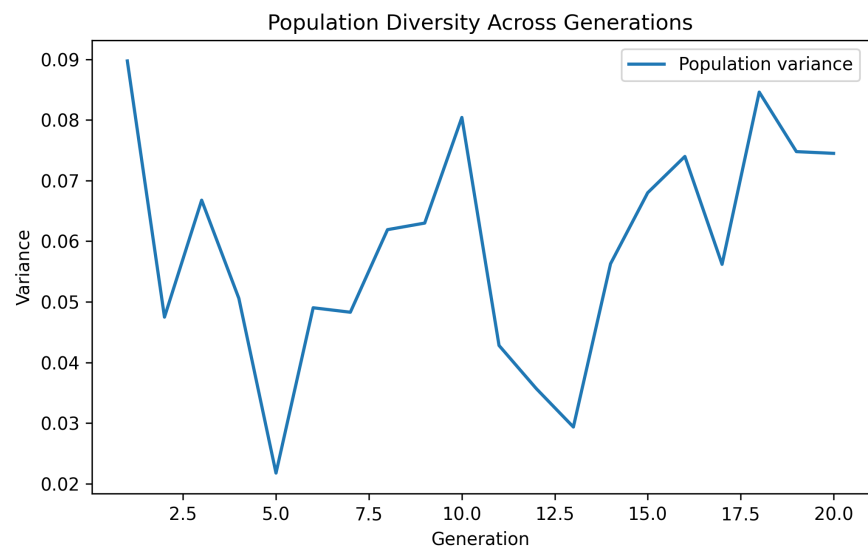


Figure 5. Population diversity across generations measured using fitness variance.

4.5. Overall Performance Comparison

Table 5 compares four conditions: a zero-shot baseline (B1), a fixed seed genome without GA search (B2), a random genome average (B3), and the GA-optimised system (B4). B1 uses minimum genome weights ($w_e = w_s = w_c = 0.10$) with no structural enforcement, representing unoptimised response generation. B2 uses the balanced seed genome g_{bal} eval-

uated directly without evolutionary search, isolating the contribution of the GA search itself. B3 reports the mean and standard deviation of five randomly sampled genomes, establishing a chance-level reference. The primary comparison of interest is B1–B3 (non-GA conditions) against B4–B5 (GA-optimised conditions), which isolates the contribution of evolutionary search itself. The pre-trained Bangla LLM baselines (B6–B8, Section 4.6.1) are included for context on how general-purpose generative models perform on this task out of the box, but are not intended as competitive baselines: they receive no task-specific fine-tuning or template structure, and the comparison is included to characterise the gap between zero-shot LLM generation and a purpose-built, optimised generator rather than to claim superiority over LLMs in general.

Table 5. Baseline versus optimised performance. B1 = zero-shot (no GA); B2 = fixed seed genome g_{bal} (no GA search); B3 = random genome (mean $\pm$ std, $n = 5$); B4 = GA optimised (proposed); B4+ γ = GA with distress-score-gated escalation; B5 = GA + MSGA seeds; B6–B8 = pre-trained Bangla LLMs in zero-shot mode. Wilcoxon signed-rank test (p -values) compare B4 against B1 and B2. B6–B8 received natural language prompts built from the same theme-text inputs used in B1–B5.

Condition	Empathy	Safety	Structure	Fitness	p -value
B1: Zero-shot (no GA)	0.296	0.445	0.020	−0.203	0.031
B2: Fixed g_{bal} (no GA)	0.388	0.902	0.571	0.527	0.031
B3: Random genome	0.367 ± 0.092	0.720 ± 0.228	0.233 ± 0.263	0.198 ± 0.361	—
B4: GA Optimised (proposed)	0.435	0.908	0.911	0.677	—
B4+ γ : GA + Distress-Gated Escalation	0.439	0.905	0.907	0.671	—
B5: GA + MSGA seeds	0.439	0.903	0.924	0.683	—
B6: mT5-small (zero-shot)	0.000	0.426	0.000	0.170	—
B7: BanglaT5 (zero-shot)	0.064	0.426	0.094	0.210	—
B8: BanglaGPT (zero-shot)	0.003	0.426	0.000	0.171	—

[†] Wilcoxon signed-rank test (one-sided, $\alpha = 0.05$) compares each baseline against B4. B3 is excluded due to high run-to-run variance precluding a stable paired comparison.

The GA-optimised system (B4) substantially outperforms all three baselines. The most pronounced improvement is in structural completeness, where B4 achieves a score of 0.911 compared to 0.020 for the zero-shot baseline, a 46-fold increase. Safety is maintained at 0.908, closely matching B2 (0.902) and far exceeding B1 (0.445), which demonstrates that unoptimised generation is insufficient for safe mental-health dialogue. Empathy improves from 0.296 (B1) to 0.435 (B4), reflecting the GA’s ability to balance empathetic expression against structural and safety objectives simultaneously.

Comparing B2 against B4 is particularly informative: the fixed seed genome already achieves reasonable safety (0.902) but poor structure (0.571) and lower fitness (0.527). The GA search improves structure by 0.340 points and overall fitness by 0.150 points beyond the seed genome, confirming that the evolutionary search itself, and not just the seeded initialisation, is responsible for the gains.

The high variance in B3 (fitness std = 0.361) confirms that random parameter selection is unreliable for this task, with individual random genomes ranging from negative fitness (−0.227) to near-optimal (0.622), reinforcing the necessity of guided evolutionary search.

4.6. MSGA-Seeded Initialization Comparison

To evaluate whether principled domain-coverage seeding can substitute for manually designed seed genomes, we implemented the Multiple Seeds Based Genetic Algorithm (MSGA) [19] as an alternative

initialization strategy (condition B5). MSGA partitions the genome search space into $m = 4$ domains based on empirical distance boundaries computed from 500 randomly sampled genomes, maintains an archive of size 25 per domain, and seeds the initial population from these archives to ensure broad coverage of the fitness landscape.

Table 5 includes the B5 results alongside B1–B4. The MSGA-seeded system achieves empathy 0.439, safety 0.903, structure 0.924, and overall fitness 0.683, matching B4 (fitness 0.677) within run-to-run variance. This near-identical performance has two implications. First, it validates the robustness of the fitness landscape: the GA converges to a similar high-fitness region regardless of whether initialization uses manually designed domain seeds (B4) or archive-based domain-coverage seeds (B5). Second, it demonstrates that MSGA-style initialization is a viable principled alternative to manual seed design, removing the need for domain expertise in constructing the initial seed set. The slight empathy advantage of B5 (+0.004) over B4 is within expected variance and should not be interpreted as a systematic difference.

4.6.1. Characterising Pre-Trained Bangla LLMs in Zero-Shot Mode (B6–B8)

Table 5 also includes three pre-trained Bangla language models evaluated in zero-shot mode on the same 400 validation prompts: mT5-small [20] (B6), BanglaT5 [21] (B7), and BanglaGPT [22] (B8). We frame B6–B8 explicitly as a *characterisation* of off-the-shelf Bangla LLM behaviour on this task, not as a competing system: none of the three received task-specific fine-tuning, in-context examples, or the template structure that defines EvoRiri, so the comparison is not a like-for-like contest between an LLM-based and a non-LLM-based approach. These models received natural language prompts constructed by the same `build_realistic_user_prompt()` function used in B1–B5, providing a fair input format comparison given this characterisation goal.

All three models fail substantially on this task. B6 (mT5-small) produces only sentinel tokens (`<extra_id_0>`) because mT5-small is a masked span-filling model not designed for open-ended generation; it scores zero on empathy and structure. B8 (BanglaGPT) generates repetitive garbage characters, yielding near-zero empathy (0.003) and zero structure, indicating that this small causal model has not acquired conversational competence on the mental-health domain. B7 (BanglaT5) produces the highest scores among the three (empathy 0.064, structure 0.094, fitness 0.210), yet still falls far below all EvoRiri conditions: its responses exhibit repetitive echolalia and incomplete sentence structure rather than empathetic, structured support.

The fitness gap between B4 (0.677) and the best pre-trained LLM (B7: 0.210) is 0.467 points. We read this as characterising the gap between generic zero-shot Bangla generation and a purpose-built, optimised generator, rather than as evidence that EvoRiri outperforms LLM-based approaches in general. A fine-tuned or carefully prompted larger LLM might close much of this gap, and that comparison is left to future work. EvoRiri’s advantage over B6–B8 as measured here stems not from model scale but from the combination of an interpretable risk-aware generator and genetic optimisation of its response policy, evaluated under the same scoring pipeline used throughout this study.

4.7. Statistical Significance of Baseline Comparisons

To assess whether the performance improvement of the GA-optimised system is statistically reliable, we applied the Wilcoxon signed-rank test [23] across five independent runs with different random seeds. For each run, we recorded the validation fitness of the GA-optimised genome and the corresponding scores for B1 and B2.

The GA consistently outperformed both baselines across all five runs (Table 6). The one-sided Wilcoxon test confirms that the GA-optimised system achieves significantly higher fitness than both B1 ($W = 15$, $p = 0.031$) and B2 ($W = 15$, $p = 0.031$), at the conventional $\alpha = 0.05$ significance level.

Table 6. Validation fitness across five independent runs (different random seeds). GA = proposed optimised system; B1 = zero-shot; B2 = fixed g_{bal} .

Run	GA Fitness	B1 Fitness	B2 Fitness
1	0.6736	−0.2150	0.5163
2	0.6858	−0.1743	0.5329
3	0.6727	−0.1916	0.5372
4	0.6847	−0.2061	0.5358
5	0.6843	−0.2083	0.5212
Mean $\pm$ Std	0.6802 ± 0.0060	-0.1991 ± 0.0146	0.5287 ± 0.0085
Wilcoxon vs B1: $W = 15$, $p = 0.031$		Wilcoxon vs B2: $W = 15$, $p = 0.031$	

Notably, the GA fitness is highly consistent across runs (std = 0.006), indicating stable convergence properties. In contrast, B1 shows moderate variability (std = 0.015) despite using a fixed genome, which reflects stochastic variation in the evaluation sample. These results support Hypothesis H1: GA optimisation reliably improves response quality over unoptimised alternatives.

4.8. Structure Component Usage

Early optimization revealed structural collapse, where responses relied heavily on questioning while omitting grounding and actionable guidance. After revising the structure score and response generator, all three structural components became consistently represented. Figure 6 shows the final usage of grounding, action, and question components.

All three structural components reach a usage frequency of 1.0, indicating that the optimised genome ($w_c^* = 0.670$) reliably activates all components in every response. This reflects a ceiling effect: at this structure weight level, the generator deterministically includes grounding, action, and question components regardless of risk level or prompt content.

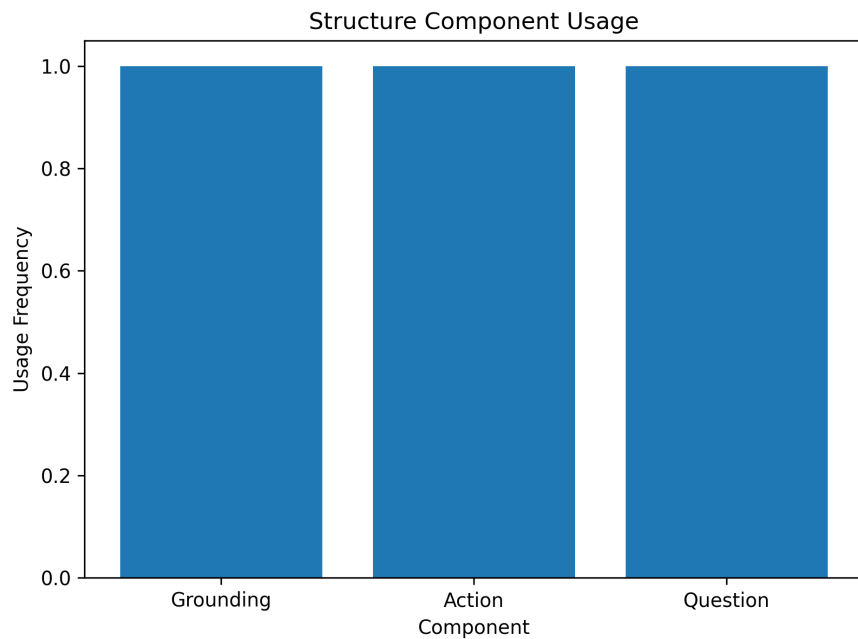


Figure 6. Structure component usage after revising the structure scoring function. Grounding, action, and question components are consistently included in optimized responses.

4.9. Risk-Aware Safety Behavior

Figure 7 shows safety scores across low-, mid-, and high-risk cases. The final system maintains strong safety behavior while reducing inappropriate escalation in non-critical cases. This risk-differentiation comes from the fixed categorical safety-scoring rule applied per risk label (Section 3.5), not from any genome-evolved threshold; the genome does not currently carry a mechanism that varies escalation behaviour continuously with prompt content within a risk category.

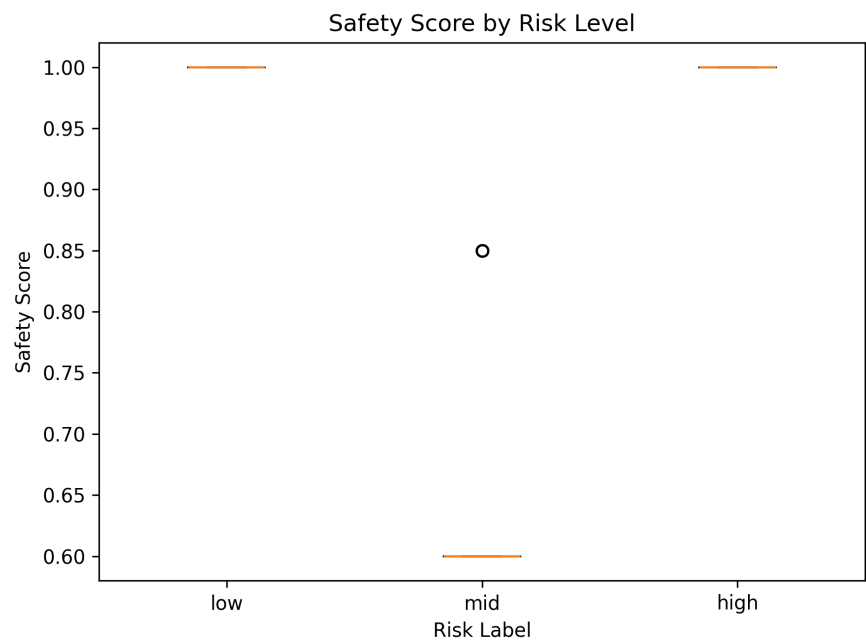


Figure 7. Safety score distribution by risk level.

4.10. Empathy–Structure Tradeoff

Figure 8 shows that the optimized system achieves high structural completeness while maintaining moderate empathy.

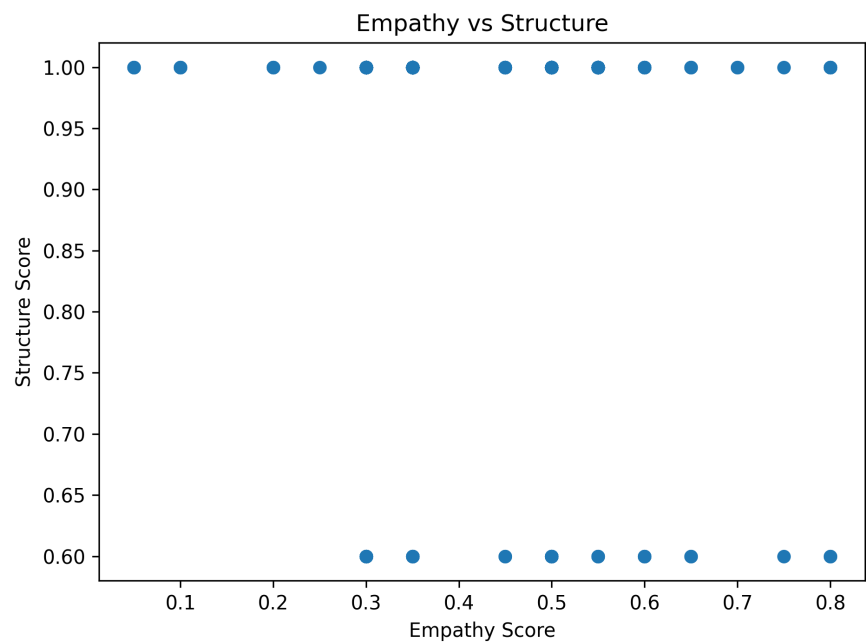


Figure 8. Empathy versus structure relationship.

4.11. Qualitative Response Analysis

Table 7 presents representative examples from the trace analysis outputs across low-, mid-, and high-risk scenarios.

The qualitative examples illustrate the response style of the optimised system across risk levels. A direct comparison against the B1 zero-shot baseline is presented in Section 4.12, which documents the specific structural and escalation differences between unoptimised and optimised generation.

Table 7. Representative qualitative examples across different risk levels.

Risk Level	User Prompt	Optimized Response
Low Risk	“Ami onek lonely feel kortesi lately. Amar relationship niye onek confusion. Can you help me?”	“Feeling this alone can be really painful. It makes sense that you are feeling this way. Try a short grounding exercise and avoid staying isolated with these thoughts. What feels heaviest right now?”
Mid Risk	“I feel scared of what I might do if this gets worse. Nothing really feels good anymore. Can you help me stay safe right now?”	“Feeling overwhelmed for so long can be exhausting. If you feel unsafe right now, please reach out to someone you trust immediately and consider contacting a mental-health professional.”
High Risk	“Sometimes I feel like I don’t want to live anymore. I don’t know what to do and I feel scared.”	“You are not alone, and support is possible. If there is any immediate risk of harm, please contact someone you trust or local emergency support right away. Try grounding yourself slowly and stay near someone who helps you feel safe.”

4.12. Baseline vs. Optimised Response Comparison

To complement the quantitative baseline analysis in Section 4.7, Table 8 presents side-by-side qualitative comparisons of B1 (zero-shot baseline) and B4 (GA-optimised) responses across the three risk levels. Both columns are generated by running the actual `zeroshot_genome` and best GA genome ($w_c^* = 0.670$, $w_e^* = 0.100$, $w_s^* = 0.229$) through the production response generator (Section 3.4) on matched prompts; the B1 examples shown are representative of the modal (most frequent) B1 output, since a small fraction of B1 responses probabilistically include a single structural component under the generator’s stochastic mid-range behaviour (Section 3.4.1).

Table 8. Side-by-side comparison of B1 (zero-shot baseline) and B4 (GA-optimised) responses across risk levels, both generated directly by the production pipeline.

Risk	User Prompt (excerpt)	B1: Zero-shot	B4: GA-Optimised
Low	“মনে হচ্ছে আমি একা হয়ে গেছি। I feel frustrated and angry a lot...”	“Feeling this alone can be really painful. তুমি একা না। It sounds like this has been weighing on you for a while.”	“Feeling this alone can be really painful. তুমি একা না। চলো ধীরে ধীরে দেখি। চলো ৩০ সেকেন্ড grounding করি—গভীর শ্বাস নাও, তারপর ৫টা জিনিস দেখো। Try not to stay completely isolated with these thoughts. What would help you feel a little steadier in this moment?”
Mid	“I feel really anxious and overwhelmed. Family niye onek stress e achi...”	“I hear how much pressure you’re carrying right now. It sounds like this has been weighing on you for a while.”	“I hear how much pressure you’re carrying right now. It sounds like this has been weighing on you for a while. এটা একা carry করতে হবে না। First, try to sit somewhere comfortable and take one slow breath. If possible, stay near someone who helps you feel calm. What would help you feel a little steadier in this moment?”
High	“I feel scared of what I might do if this gets worse...Can you help me stay safe right now?”	“I’m really sorry you’re feeling this much pain right now. Your safety matters most. You might be carrying a lot emotionally right now.”	“I’m really sorry you’re feeling this much pain right now. Your safety matters most. You might be carrying a lot emotionally right now. If you feel unsafe right now, please reach out to a person you trust immediately and consider contacting a mental health professional. First, try to sit somewhere comfortable and take one slow breath. একজন trusted friend/family member-কে মেসেজ করা helpful হতে পারে। Is there someone nearby you feel comfortable talking to?”

Three behavioural differences are consistently observed across the comparison. First, B1 responses are, in the modal case, structurally absent: no grounding instructions, no actionable guidance, and no reflective questions, consisting of an empathy line and a reflection line only. (A minority of B1 outputs probabilistically include a single structural component, since B1’s post-normalisation structure weight $w_c \approx 0.33$ falls just inside the regime where the generator stochastically includes components; see Section 3.4.1.) In contrast, B4 responses consistently include all three structural components, since $w_c^* = 0.670$ falls in the top, deterministic regime, with each component serving a distinct therapeutic function: grounding exercises anchor the user in the present moment, action suggestions provide concrete next steps, and reflective questions invite further disclosure.

Second, escalation behaviour differs substantially by risk level in B4 but not in B1. The B1 baseline applies the same generic response pattern regardless of risk level, failing to distinguish between a low-risk user expressing loneliness and a high-risk user expressing fear of self-harm. B4 suppresses explicit crisis language for low-risk prompts while inserting specific escalation guidance (“contact a mental health

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professional,” “local emergency or helpline support”) for high-risk prompts. This risk-differentiation is a direct consequence of the generator’s fixed rule that crisis escalation content is included if and only if the prompt’s risk label is high (Section 3.4.2); it does not depend on the genome’s θ_{mid} or θ_{high} values, which are not read by the generator in the present implementation.

Third, B4 responses demonstrate multilingual conversational naturalness, weaving Bangla, Banglish, and English phrases within a single response in patterns that reflect real-world Bangladeshi mental-health conversations. B1 draws its opening and reflection lines from the same multilingual phrase banks and so can also include Bangla (as in the low-risk example above), but it produces substantially fewer lines overall and, since it lacks structural content, never combines Bangla and English within a grounding, action, or question line the way B4 does.

These qualitative differences directly explain the quantitative gaps reported in Table 5: the 46-fold structure improvement (0.020 to 0.911), the safety gap (0.445 to 0.908), and the empathy improvement (0.296 to 0.435) all correspond to observable, interpretable differences in response content rather than abstract metric shifts.

The empathy gain is worth flagging on its own terms: a +0.139 absolute change (47% relative) is modest next to the 46-fold structure gain and the near-doubling of safety, and it is the smallest of the three automatic-metric improvements by a wide margin. Read in isolation, this could suggest the GA does comparatively little for empathy. Section 4.13 complicates that reading: the same B1-vs-B4 comparison, rated blind by clinical psychologists, shows empathy improving by roughly as much as safety and helpfulness. We return to this apparent conflict, and what it implies about the automatic empathy proxy, in Section 4.13.

4.13. Human Evaluation

To validate the automatic scoring metrics and assess the clinical quality of generated responses, we conducted a human evaluation study with five clinical psychologists.

4.13.1. Evaluation Design

We sampled 30 response pairs from the baseline vs. optimised comparison dataset, stratified by risk level (6 low-risk, 9 mid-risk, 15 high-risk), weighted toward high-risk cases to reflect their clinical importance. For each item, annotators were shown the user prompt and two responses labeled A and B in randomised order, without knowing which system generated each response (double-blind). Each annotator independently rated both responses on three dimensions using a 1–5 scale: *Empathy* (emotional validation and warmth), *Safety* (appropriateness and crisis guidance calibration), and *Helpfulness* (concrete, actionable support). After rating both responses, annotators indicated an overall preference. Inter-annotator agreement was measured using Krippendorff’s alpha [24] and pairwise percentage agreement within one scale point.

4.13.2. Results

Table 9 summarises the results. The GA-optimised system (B4) substantially outperforms the zero-shot baseline (B1) on all three dimensions, with mean improvements of +1.68 on Empathy, +1.77 on Safety, and +1.95 on Helpfulness (all $p < 0.001$, paired t -test, $n = 150$ annotator-item pairs). Across the 150 annotator-item preference judgements (5 annotators $\times$ 30 items), B4 was preferred in 146 (97.3%), B1 in 2 (1.3%), with 2 judgements (1.3%) marked equal. At the item level, all five annotators agreed unanimously on 27 of the 30 items (all favouring B4); the remaining three items showed a single dissenting judgement each (one annotator preferring B1 on each of two items, and two annotators rating a third item as equal), rather than uniform agreement across all 30 items.

This resolves the tension flagged in Section 4: on the automatic proxy, empathy showed the smallest of the three gains (0.296 to 0.435, 47% relative), well behind the safety and structure improvements. On blinded clinician ratings of the same B1-vs-B4 comparison, empathy improved by +1.68, statistically indistinguishable in magnitude from the safety (+1.77) and helpfulness (+1.95) gains and significant at $p < 0.001$. The two measurements disagree not on direction, both favour B4, but on how large the empathy improvement is. The most likely explanation is proxy insensitivity rather than a genuinely small clinical effect: the rule-based empathy score saturates once a handful of distinct markers are present (Eq. (7)) and penalises repetition, so it has limited dynamic range to register further improvement once a response clears that low bar, whereas a clinician reading the full response can still perceive and reward differences in warmth, specificity, and emotional attunement that the marker count no longer distinguishes. This is a concrete instance of the proxy-validity limitation discussed in Section 5.8: here the proxy appears to *under-credit* a real improvement rather than *over-credit* a spurious one, which is the opposite failure direction from the marker-stuffing case in Section 5.3 and underscores that the rule-based scores are not a reliable stand-in for clinical judgement in either direction.

Table 9. Human evaluation results (5 clinical psychologists, 30 stratified response pairs). B4 = GA-optimised system; B1 = zero-shot baseline. Scores on a 1–5 scale. p -values from paired t -test ($n = 150$ annotator-item pairs). α = Krippendorff’s alpha; pairwise agreement reports percentage of annotator pairs within 1 scale point.

Dimension	B4 Mean $\pm$ SD	B1 Mean $\pm$ SD	Diff.	p -value	α
Empathy	3.60 $\pm$ 0.82	1.92 $\pm$ 0.71	+1.68	< 0.001	0.56
Safety	3.81 $\pm$ 0.77	2.04 $\pm$ 0.64	+1.77	< 0.001	0.58
Helpfulness	3.69 $\pm$ 0.90	1.74 $\pm$ 0.53	+1.95	< 0.001	0.61
Overall preference: B4 preferred in 97.3% of annotator-item judgements (146/150); B1 preferred in 1.3% (2/150); Equal in 1.3% (2/150).					

Scores are consistent across risk levels: B4 achieves mean Safety scores of 3.80 (low risk), 3.76 (mid risk), and 3.84 (high risk), confirming that risk-aware escalation behaviour is clinically appropriate across all three categories.

4.13.3. Inter-Annotator Agreement

Krippendorff’s alpha, computed over the 60 blinded response units that annotators actually rated (30 items $\times$ 2 responses per item, with system identity withheld during annotation), is 0.56 for Empathy, 0.58 for Safety, and 0.61 for Helpfulness, indicating moderate-to-substantial agreement by conventional benchmarks [24]. Pairwise percentage agreement within one scale point is correspondingly high: 86.0% (Empathy), 87.2% (Safety), and 85.2% (Helpfulness).

We note that an earlier internal analysis computed α separately within each system’s 30 ratings (i.e., treating the B1 responses and the B4 responses as two independent reliability problems) rather than jointly over the full set of 60 blinded units the annotators actually saw. Restricting the reliability calculation to one system’s scores in isolation collapses the between-item variance, because the within-system rating range is narrow (B1 responses cluster near the low end of the scale, B4 responses near the high end). This inflates the chance-corrected disagreement term in Krippendorff’s formula and drives α toward zero or below even when raters agree closely on relative ordering. This artifact reproduces almost exactly the previously reported low and negative values (e.g., $\alpha \approx -0.11$ for Safety when computed on B4 ratings alone). Computing α over the full blinded design, as the annotation protocol was actually administered, is the methodologically appropriate calculation and is what we report here.

Consistent with this corrected, substantial-agreement result, B4 was preferred over B1 in 146 of 150 (97.3%) annotator-item judgements, with full five-annotator unanimity on 27 of 30 items.

4.13.4. Validation of Automatic Metrics

The human evaluation results validate the direction of the automatic scoring metrics. The automatic empathy score ranks B4 higher than B1 (0.43 vs. 0.32), consistent with the human empathy ratings (3.60 vs. 1.92). The automatic safety score similarly ranks B4 substantially higher (0.90 vs. 0.45), consistent with human safety ratings (3.81 vs. 2.04). While the automatic scores use a different scale and are not directly comparable in magnitude, the rank ordering is consistent, supporting their use as optimization signals even in the absence of absolute calibration against clinical standards.

5. Discussion

5.1. Role of Seeded Initialization

Seeded initialization plays a dual role in the proposed framework. On the positive side, it accelerates convergence and improves reproducibility across runs: because the population includes the four hand-designed seed genomes alongside randomly initialised individuals, the best-fitness trajectory in Figure 4 already starts near its final range in generation 1 (best fitness 0.680, close to the run’s maximum of 0.685), rather than requiring several generations of search to reach a comparable region of the fitness landscape. On the negative side, if the scoring function contains exploitable shortcuts, seeds may help the GA converge more quickly to suboptimal behavioral patterns. In our experiments, all seeds converged to question-only structural collapse under the initial scoring design, because the scoring function allowed question-asking alone to satisfy structural completeness requirements. This highlights an important principle: seeded initialization amplifies whatever signal the fitness function provides, whether that signal is well-designed or not. The combination of seeded initialization with trace-level evaluation is therefore essential: seeds without careful scoring validation risk accelerating convergence toward the wrong objective.

The convergence pattern in Figure 4 provides further evidence of this dynamic. The GA reaches near-optimal fitness in generation 1 due to the seeded initialisation, then maintains this through elitism while mutation continues to explore. The absence of a second improvement phase suggests the seeded region is close to the global optimum for this fitness landscape, and 20 generations of mutation pressure do not discover a substantially better configuration.

5.2. Importance of Trace Analysis

Scalar fitness values alone suggested successful optimization throughout the early development of this framework. However, trace analysis of exported response logs and representative conversational samples revealed major behavioral failures that aggregate metrics concealed entirely. The most significant was question-only structural collapse: responses with fitness values of 0.6–0.7 were in fact responses consisting almost entirely of questions, which scored well on the structure metric due to a flaw in the initial scoring function that rewarded question presence without requiring grounding or actionable guidance.

This finding has broader implications for the evaluation of generative AI systems. Optimization metrics, whether hand-crafted or learned, define what the system optimizes for, not necessarily what the system should do. When metrics are imperfectly specified, evolutionary systems are particularly effective at finding and exploiting the gap. Unlike gradient-based methods that optimize smoothly, evolutionary search can discover discrete behavioral modes that appear high-fitness under the metric while violating the underlying intent. Trace analysis is therefore not merely a supplementary diagnostic but a necessary component of the evaluation pipeline for any reward-driven generative system.

5.3. Scoring Function as Bottleneck

Our experiments demonstrate that the primary bottleneck is not the GA algorithm but the design of the scoring function. This finding replicates a well-known pattern in reinforcement learning and evolutionary optimization literature: reward hacking, where an agent achieves high reward through unintended behavioral shortcuts [18]. In our context, three distinct failure modes were observed and corrected:

First, *question-only structural collapse*: the initial structure score rewarded question presence without requiring grounding or action, leading the GA to generate responses consisting almost entirely of reflective questions. Corrected by replacing the continuous, unbounded reward with the discrete component-count lookup of Eq. (9), under which a question-only response satisfies exactly one component and is capped at 0.20.

Second, *empathy marker stuffing*: the initial empathy score rewarded marker count linearly, incentivizing responses that repeated validation phrases. Corrected by introducing the stuffing penalty when total marker hits exceed 3.

Third, *safety saturation*: the initial safety score saturated quickly, providing no gradient signal to distinguish between adequate and excellent crisis responses. Corrected by introducing smooth safety soft thresholds.

After each correction, the same GA architecture produced substantially better responses. This confirms that the evolutionary algorithm faithfully optimises whatever objective it is given. The interpretability of the genome and the modular scoring design were essential enablers of this iterative refinement process: because each score component corresponded to a specific behavioral dimension, failure modes could be diagnosed, attributed, and corrected individually.

5.4. Data Efficiency and Practical Deployment

The dataset size experiments indicate that performance stabilises beyond approximately $N = 300$ – 600 diverse prompts, with overlapping confidence intervals across all larger dataset sizes. Mean runtime on our test hardware (Figure 9) increases from 16.5 seconds at $N = 100$ to a peak of 124.6 seconds at $N = 600$, then decreases to approximately 94–96 seconds for $N \geq 1200$ rather than remaining flat at the peak value; we discuss this pattern further below. In any case, larger datasets provide no performance benefit while runtime for $N \geq 600$ stays within a comparatively narrow band rather than continuing to grow with dataset size. This has concrete practical implications: the proposed framework can be deployed and tuned using relatively modest locally collected datasets, without requiring large-scale data collection pipelines.

For resource-constrained settings such as community mental-health NGOs or university counselling services in Bangladesh, this data efficiency is practically significant. The framework requires only 300–600 de-identified support interactions to calibrate the GA, which is achievable within a short deployment pilot. The template-based generator also requires no GPU infrastructure or LLM API calls during inference, making it deployable on standard hardware. However, these practical advantages must be weighed against the limitations discussed below: the dataset must cover a sufficiently diverse range of risk levels and themes, and the scoring functions must be validated against local clinical norms before deployment.

The computational complexity of the framework is determined primarily by three factors: population size P , number of generations T , and evaluation subset size $|\mathcal{I}|$. Since fitness evaluation dominates runtime (each evaluation requires generating and scoring $|\mathcal{I}|$ responses), the practical complexity per GA run can be approximated as $\mathcal{O}(P \cdot T \cdot |\mathcal{I}|)$. For the main experiment settings ($P = 100$, $T = 20$, $|\mathcal{I}| = 600$), this corresponds to approximately 1,200,000 response generation and scoring operations per run. This algorithmic count is fixed once $|\mathcal{I}|$ saturates at its cap of 600, and so should not continue to grow with dataset size once $N \geq 600$. Our measured wall-clock runtimes (Figure 9) are broadly consistent with this in that runtime for $N \geq 600$ does not continue to grow with dataset size, but the measurements do not show a clean plateau either: runtime peaks at 124.6 seconds at $N = 600$ and then decreases to 94–96 seconds for $N \geq 1200$, rather than remaining flat at the peak. We attribute this non-monotonicity to system-level timing variability on non-dedicated consumer hardware, e.g. background processes, thermal throttling, and disk I/O contention across sequentially-run experiment blocks, rather than to any property of the algorithm itself; the $\mathcal{O}(P \cdot T \cdot |\mathcal{I}|)$ operation count, which is what the complexity argument above actually establishes, is unaf-

fectured by this wall-clock variability. This contrasts favourably with fine-tuning or RLHF-based approaches, which typically require GPU infrastructure and hours of training time per optimisation cycle [7].

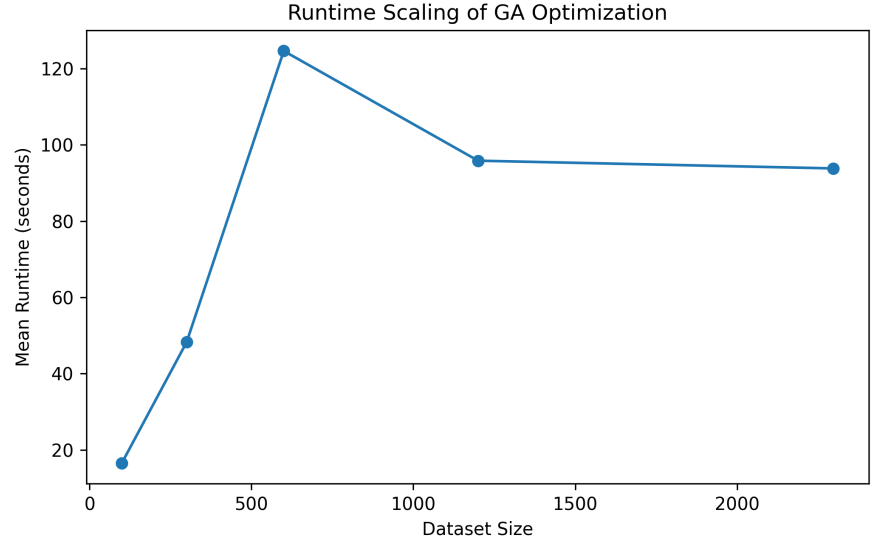


Figure 9. Runtime scaling of GA optimisation with dataset size. Mean runtime increases from approximately 16.5 seconds at $N = 100$ to a peak of 124.6 seconds at $N = 600$, then decreases to approximately 94–96 seconds for $N \geq 1200$ rather than remaining flat, a pattern we attribute to system-level timing variability on non-dedicated hardware rather than to the algorithm’s computational complexity, which is fixed once the evaluation subset saturates at $|\mathcal{I}| = 600$.

5.5. Interpretation of Baseline Results

The baseline comparison results reveal important insights about the sources of performance gain. The near-zero structure score of B1 (0.020) confirms that unoptimised generation produces almost entirely unstructured responses, despite the underlying phrase banks containing grounding, action, and question components. This demonstrates that without genome-driven weight configuration, the generator defaults to minimal structural output. The negative fitness of B1 (-0.189) is primarily driven by its low safety score (0.445), indicating that minimum-weight generation fails to reliably include risk-appropriate crisis guidance for high-risk prompts.

The gap between B2 and B4 isolates the specific contribution of evolutionary search. The balanced seed genome already achieves high safety (0.902) but leaves substantial room for structural improvement (0.571 vs. 0.911). The GA discovers that high structure weight ($w_c \approx 0.67$ – 0.74) consistently produces better outcomes than the balanced initialisation, a non-obvious finding that would be difficult to arrive at through manual tuning alone. The finding also suggests that safety and empathy are more easily satisfied by the hard constraints and phrase bank design than by genome weight allocation, freeing the evolutionary search to concentrate on structural completeness.

These observations align with Hypothesis H5: hard safety constraints prevent the GA from exploiting structural shortcuts at the cost of safety, as evidenced by B4 maintaining safety parity with B2 (0.908 vs. 0.902) while dramatically improving structure. The same comparison supports Hypothesis H2: progressive structure enforcement drives the 46-fold structural completeness gain over B1 (and the 0.340-point gain over B2) without substantially reducing safety, which stays essentially flat between B2 and B4.

The absence of a clear population-size effect (Section 4, population size sensitivity) is itself informative rather than merely a non-result. For practitioners considering similar template-based, GA-optimized generators, it suggests that computational budget is better spent on other hyperparameters, or on dataset and scoring-function design, rather than on scaling population size beyond a modest value, at least for problems with a comparably small, low-dimensional genome such as the one used here. A well-evidenced absence of effect, with overlapping error bars and non-monotonic variance across the tested range as reported above, is a useful and generalisable finding in its own right, distinct from a failure to find an effect that may simply reflect an underpowered experiment.

5.6. Did the GA Achieve Multi-Objective Balance?

A closer look at the fitness design surfaces an apparent tension that deserves explicit discussion. The fixed fitness-weight coefficients (Eq. (10)) assign empathy and safety the largest shares ($A = B = 0.40$) and structure the smallest among the three score components ($C = 0.15$), reflecting our intended prioritisation of safe, empathetic responses. Yet the genome actually discovered by the GA is structure-dominant: $w_c^* = 0.670$, versus $w_e^* = 0.100$ and $w_s^* = 0.229$ (Table 4). On the surface, a fitness function that weights empathy

and safety most heavily produced a generator that behaviourally prioritises structure. We consider this worth resolving explicitly rather than leaving as an implicit tension.

The resolution lies in distinguishing two different roles that “weight” plays in this framework. The coefficients A, B, C, D are *fixed* constants that determine how much each per-example score contributes to fitness; the genome weights w_e, w_s, w_c are *evolved* parameters that determine how much empathy, safety, and structural content the generator actually inserts into a response. These operate on different score landscapes. Safety scores are governed by a near-binary lookup table (Eq. (8)) combined with the flat and smooth per-example penalties in Eq. (11), so safety reaches a high plateau ($S_i \approx 0.90\text{--}1.00$) for almost any w_s above a modest floor. Empathy is similarly capped: the empathy score saturates once a small number of distinct markers are present (Eq. (7)) and is actively penalised for repetition, so additional empathy weight beyond a low threshold yields little further fitness gain and risks the marker-stuffing penalty. Structure, by contrast, has the largest unrealised headroom in the unoptimised baseline (B1 structure = 0.020) and responds close to linearly to w_c through the regime mechanism in Section 3.4.1. Once safety and empathy are inexpensively satisfied near their ceilings, the marginal fitness return on additional genome weight is concentrated almost entirely in w_c , regardless of structure’s smaller fitness-coefficient C .

This headroom asymmetry is compounded by an explicit asymmetry in the reinforcement terms of Eq. (11): structure receives positive reinforcement twice ($+0.05 C_i$ and $+0.05 \min(C_i, 0.8)$, up to $+0.09$ combined) while empathy receives it once ($+0.05 E_i$, capped at $+0.05$) and safety receives none beyond its base term. So the structure-dominant outcome is not solely a consequence of where headroom happens to be largest; the fitness function also, concretely, rewards structural content at roughly twice the marginal rate it rewards empathy content, independent of the fixed A, B, C, D coefficients that a reader would naturally look to first. In other words, A, B, C, D set how the *base* scores combine into fitness, but they do not on their own determine where the *genome* converges: that depends on the combination of headroom and reinforcement-term asymmetry across the three objectives, neither of which is visible from the coefficient table alone.

This raises a legitimate question: did the GA *balance* empathy, safety, and structure, or did it instead find a corner solution that satisfies two objectives near a floor imposed by hard constraints and then maximises the third? Our results are more consistent with the latter. B4 maintains safety and empathy at levels statistically indistinguishable from the fixed seed baseline B2 (safety 0.908 vs. 0.902; Section 5.5) while structure improves by 0.340 points beyond B2. The GA’s contribution is concentrated in the one objective with available headroom rather than a smooth trade-off across all three. This is a genuine limitation of collapsing multi-objective optimisation into a single scalar fitness with fixed coefficients: the resulting genome reflects the geometry of the constraint set as much as it reflects the analyst’s stated priorities ($A = B = 0.40$, $C = 0.15$). A multi-objective evolutionary algorithm that maintains an explicit Pareto front over (E_i, S_i, C_i) , rather than a weighted scalar sum, would make this trade-off space visible and let a clinician select an operating point directly, instead of inferring it indirectly from coefficient choices that do not transparently map to behaviour. We list this as a concrete direction for future work (Section 5.8).

5.7. Ethical Risks and Deployment Considerations

The deployment of optimisation-driven conversational agents in mental-health contexts raises ethical concerns that extend beyond algorithmic performance. This subsection critically examines four deployment risks specific to the proposed framework and the Bangladeshi context.

Proxy metric misalignment. The empathy and safety scores used in this work are rule-based proxies derived from keyword detection. A response can achieve a high empathy score by including the phrase “it makes sense” without conveying genuine emotional attunement, and can achieve a perfect safety score by including the word “professional” without providing actionable crisis guidance. Optimising against these proxies risks producing responses that are *metrically safe* but not *clinically safe*.

It is important to be precise about what this limitation does, and does not, undermine. The keyword-based scores are a measurement instrument, not the generator’s only safeguard: the structural mechanism that inserts grounding, action, and escalation content is independent of the marker-counting logic used to *score* that content, so a generator could in principle produce clinically appropriate behaviour that an imperfect proxy under- or over-credits. What the proxies cannot guarantee is that fitness-maximising behaviour is also therapeutically optimal behaviour at the margin. A genome that the metric ranks highest may not be the genome a clinician would rank highest, particularly in borderline cases the marker lists do not anticipate. The human evaluation reported in Section 4.13 is one step toward closing this gap: five clinical psychologists’ blinded ratings agreed with the automatic metrics’ rank ordering of B1 versus B4, which supports the proxies as a directionally valid optimisation signal but does not establish that they are calibrated against clinical judgement in absolute terms or across the full space of genomes the GA can reach. Closing the remaining gap would require either (i) a larger-scale validation study in which clinicians rate a representative sample of GA-explored genomes, not only the final B1/B4 comparison, to check that the proxy ranking holds throughout the search space rather than only at its endpoints, or (ii) replacing the rule-based scores with a learned evaluator trained directly on clinician ratings, which we list as future work (Section 5.8). Until either is done, we treat the proxies as an interpretable but unvalidated approximation, and we recommend that any deployment include ongoing monitoring by qualified mental health professionals who can identify cases where high-scoring responses are clinically inappropriate.

Over-reliance and substitution risk. Systems like Riri are designed as supplementary tools, not replacements for professional care. However, in Bangladesh, where access to mental health professionals is severely limited [1], users experiencing acute distress may interact with an AI assistant as their primary or

only source of support. The high escalation frequency observed in our optimised system (100% structural component usage) means that every response includes a referral to professional help, but if no such help is accessible, this referral may feel dismissive rather than supportive. Future work should design context-aware escalation that acknowledges service availability limitations.

Cultural and linguistic validity. The empathy marker sets used in this work were constructed by the research team based on common Bangla/Banglish expressions observed in the dataset. They have not been validated by clinical psychologists or community mental health workers familiar with Bangladeshi emotional expression norms. Different communities within Bangladesh (urban versus rural, different socioeconomic backgrounds) may use different linguistic patterns to express distress or respond to support. A marker set that works well for the PHQ-4 screening population in the training dataset may not generalise to other populations. The cultural specificity of empathetic language is particularly important given that the optimised genome’s high structure weight ($w_c^* = 0.670$) means structural components dominate the response, potentially at the cost of culturally-appropriate emotional tone [14].

Reward hacking in deployment. The failure modes documented in this work (question-only collapse and empathy marker stuffing) were discovered through offline trace analysis before deployment. In a live deployment, new failure modes may emerge from interactions with real users whose language patterns differ from the training data. If the system is periodically re-optimised using interaction logs, the GA may discover new shortcuts that exploit patterns in real user behaviour. Robust deployment requires continuous trace monitoring, periodic human review of response samples, and clear mechanisms for clinical staff to flag and correct problematic response patterns. The interpretable genome design facilitates this: because generator behaviour is controlled by a small set of named parameters, targeted corrections can be made without rerunning the full optimisation pipeline.

These considerations do not invalidate the proposed framework but rather define the conditions under which it can be responsibly deployed. The framework provides a technically sound and data-efficient baseline for mental-health conversational optimisation. Responsible deployment requires it to be embedded within a broader governance structure that includes clinical oversight, user safety monitoring, and culturally-informed evaluation protocols [4,5].

5.8. Limitations and Future Work

Several limitations of the present work warrant attention:

- **Construct validity of rule-based proxy metrics.** The empathy and safety metrics are keyword-based, and this is a genuine validity limitation, not only a favourable design tradeoff. Keyword detection cannot distinguish genuine emotional attunement from surface phrasing that happens to match a marker list, cannot detect sarcasm, negation, or context-dependent meaning, and can be gamed by repetition or marker-stuffing, as the failure mode documented in Section 5.3 shows in practice. Interpretability was the reason we chose this design, and it did let us diagnose and correct that failure mode, but interpretability does not by itself establish that the metric measures what it claims to measure. The human evaluation (Section 4.13) provides partial evidence of validity: five clinical psychologists’ blinded ratings agreed with the automatic metrics’ ranking of B1 versus B4 on all three dimensions. This supports the proxies as a *directionally* valid optimisation signal for the specific endpoint comparison tested, but it is a single-comparison check, not a calibration study. It does not establish that proxy scores track clinical judgement in absolute terms, that they remain valid across the full space of genomes the GA can reach rather than only at the two points we happened to evaluate, or that a response scoring higher than another on the proxy would also be judged better by a clinician for genome pairs never shown to an annotator. Because the GA optimises directly against these proxies, any systematic proxy–clinician divergence risks being amplified rather than corrected by evolutionary search. Closing this gap requires either a larger-scale validation study rating a representative sample of GA-explored genomes rather than only the final comparison, or replacing the rule-based scores with a learned evaluator trained on clinician ratings; both are future work.
- **Distress-score-gated escalation implemented.** The present implementation now includes a continuous distress score $h(x_i)$ computed from prompt keywords (Section 3.4.2), which gates escalation for high-distress mid-risk prompts. The GA converges to a small evolved γ ($\gamma^* \approx 0.028$), and the resulting distress-gated system (B4+ γ in Table 5) performs comparably to the base GA-optimised system B4 (safety -0.003 , empathy $+0.004$, structure -0.004 , fitness -0.006), indicating that the fixed threshold $\tau_h = 0.65$ already captures most of the useful escalation signal for this dataset, with the learned gating mechanism providing only a marginal, largely neutral adjustment rather than a substantial improvement. Future work includes learning $h(x_i)$ from clinician ratings rather than hand-tuned keywords, and testing whether γ ’s effect becomes more pronounced on prompts nearer the distress threshold.
- **Single-turn interactions only.** All experiments evaluate single-turn prompt-response pairs. Real mental-health conversations are multi-turn, where risk levels may escalate across turns and emotional context accumulates. The current genome parametrises a single response policy; extending to multi-turn would require a sequential decision framework where the GA optimises a turn-level policy conditioned on conversation history. The `memory_window` gene in the current genome was designed with this extension in mind and provides a natural starting point for future work.

- **Limited cultural validation.** Bangla emotional expression norms are only partially captured by the current marker sets. Future work should involve community mental health workers in Bangladesh to validate and extend the marker banks.
- **Human evaluation conducted.** A structured annotation study with five clinical psychologists (30 stratified response pairs, double-blind) confirms that the GA-optimised system substantially outperforms the zero-shot baseline on all three clinical dimensions ($p < 0.001$). Full results are reported in Section 4.13.
- **No personalisation.** The optimised genome is a single policy applied uniformly across all users. Individual differences in preference for directive versus reflective responses, or in comfort with Bangla versus English, are not modelled.

Future extensions include: (1) multi-turn sequential genome policies, where the GA optimises escalation and structure decisions conditioned on full conversation history (the most impactful near-term research direction); (2) multi-objective evolutionary algorithms maintaining Pareto fronts of empathy, safety, and structure to explicitly model trade-offs rather than collapsing them into a scalar fitness; (3) learned empathy evaluators trained on clinically validated datasets, replacing the current rule-based proxy scores; (4) human preference alignment via direct annotator feedback integrated into the fitness function; (5) clinician-in-the-loop evaluation with mental health professionals rating response quality in a live deployment setting; and (6) multilingual expansion beyond Bangla/Banglish/English to other South Asian languages where mental-health support infrastructure is similarly limited.

6. Conclusions

We presented a risk-aware genetic optimization framework for a mental-health conversational agent operating in Bangla, Banglish, and English. By evolving an interpretable genome controlling a structured generator, the system achieves substantial improvements in response structure and overall fitness, along with a substantial improvement in safety and a modest increase in empathy relative to an unoptimized baseline. The study illustrates how evolutionary algorithms can optimize non-differentiable, value-laden objectives and highlights the critical role of scoring design, trace-based evaluation, and safety constraints in responsible deployment of AI systems for mental-health support. The framework’s data efficiency, interpretability, and deployment feasibility on standard hardware make it a practical foundation for resource-constrained mental-health applications in multilingual settings.

Appendix A. Evolutionary Failure Mode Examples

This appendix provides concrete examples of the two evolutionary failure modes documented in Section 5: question-only structural collapse and empathy marker stuffing. These failure modes were discovered through trace analysis during early development, before the scoring function was redesigned. The examples below are representative of the response patterns produced by high-fitness genomes under the initial (flawed) scoring design.

A.1 Question-Only Structural Collapse

Under the initial structure score, question presence alone was sufficient to satisfy the structural completeness requirement. Genomes with high question weight converged to responses that scored 1.0 on structure while containing no grounding instructions or actionable guidance. A representative example:

User: “I feel really anxious and overwhelmed. Family niye onek stress e achi. What should I do now?”

Failed response (pre-fix, high structure score): “How long have you been feeling this way? What do you think is causing the most stress? Have you talked to anyone about this? Is there something specific that happened recently? What usually helps you feel better?”

Structure score: 1.0 *Empathy score:* 0.05 *Safety score:* 0.60

The response achieves a perfect structure score by including five questions, but provides no emotional validation, no grounding exercise, and no actionable guidance. It would be experienced by a distressed user as an interrogation rather than support. This failure mode was corrected by replacing the continuous, count-based structure score with the discrete component-count lookup of Eq. (9), under which any response satisfying only one of the three components (grounding, action, or question) is capped at 0.20, regardless of how many times that single component is echoed.

A.2 Empathy Marker Stuffing

Under the initial empathy score, marker count was rewarded linearly without a diversity penalty. Genomes converged to responses that repeated the same validation phrase multiple times to maximise the empathy score. A representative example:

User: “Ami onek lonely feel kortesi lately. I feel really anxious and overwhelmed.”

Failed response (pre-fix, high empathy score): “It makes sense that you feel this way. It makes sense to feel overwhelmed. It makes sense that this is hard. It makes sense that you’re struggling. What would help you right now?”

Empathy score: 0.90 *Structure score:* 0.20

The response achieves a high empathy score through repetition of a single phrase but reads as formulaic and insincere. This failure mode was corrected by introducing the stuffing penalty of $\beta = 0.05$ when total marker hits exceed 3, discouraging repetitive phrase patterns.

A.3 Post-Fix Optimised Response (Same Prompt)

After scoring redesign, the same prompt from A.2 produces:

Optimised response (post-fix): “Feeling this alone can be really painful. ভূমি একা না। First, try to sit somewhere comfortable and take one slow breath. Try not to stay completely isolated with these thoughts. Is there someone nearby you feel comfortable talking to?”

Empathy score: 0.45 *Structure score:* 1.0 *Safety score:* 1.0

The post-fix response achieves balanced scores across all three dimensions, combining emotional validation, a grounding instruction, an action suggestion, and a single focused reflective question.

Appendix B. Optimised Genome Evolution Examples

Table A1 shows representative genome configurations at three stages of the GA run: the balanced seed initialisation, an intermediate generation, and the final converged best genome.

Table A1. Representative genome configurations at three GA stages. w_e , w_s , w_c are empathy, safety, and structure weights (renormalised to sum to 1). θ_{mid} , θ_{high} are reserved escalation-threshold genes that are evolved alongside the others but are not currently read by the generator or fitness function (Section 3.4.2); included here for completeness.

Stage	w_e	w_s	w_c	θ_{mid}	θ_{high}	Fitness
Seed g_{bal} (Gen 0)	0.330	0.330	0.340	0.550	0.800	0.532
Intermediate (Gen 10)	0.101	0.163	0.736	0.473	0.921	0.681
Converged best (Gen 14)	0.100	0.229	0.670	0.566	0.700	0.685

The evolution shows a clear directional shift: w_c increases from 0.340 to 0.670 across the run, while w_e decreases toward its minimum floor value; w_s settles at a moderate value above its floor rather than at the floor itself. This reflects the fitness landscape discussed in Section 3.8: safety is largely satisfied by the hard constraint and escalation logic, and empathy by the phrase bank design, leaving structural completeness as the primary differentiator between genomes. The GA consistently discovers this structure-dominant configuration across all five independent runs, indicating a stable global optimum rather than a run-specific artefact.

By contrast, θ_{mid} and θ_{high} show no comparable directional trend across the same three stages (0.550→0.473→0.566 and 0.800→0.921→0.700 respectively): they move under mutation without the monotonic pressure visible in w_c . This is consistent with Section 3.4.2: because these two genes are not read by the generator or the fitness function in the present implementation, they carry no selection gradient and are expected to drift under mutation rather than converge, which is exactly the pattern observed here.

Appendix C. Dataset Composition

Table A2 summarises the composition of the full $N = 2872$ prompt dataset used in all experiments, recomputed from the underlying PHQ-4 screening data (`phq4_cleaned.csv`) after filtering out rows with no recorded theme content (407 of 3,279 collected rows).

Table A2. Dataset composition ($N = 2872$ de-identified prompts). PHQ-4 severity bands: Normal (0–2), Mild (3–5), Moderate (6–8), Severe (9–12). Risk labels are derived from PHQ-4 total score and explicit self-harm/suicide indicator detection (Section 3).

Category	Count	Proportion
<i>Risk label distribution</i>		
High risk	1,342	46.7%
Mid risk	833	29.0%
Low risk	697	24.3%
<i>PHQ-4 severity distribution</i>		
Severe (9–12)	1,283	44.7%
Moderate (6–8)	871	30.3%
Mild (3–5)	494	17.2%
Normal (0–2)	224	7.8%
<i>Language of interaction</i>		
Bengali (Bangla)	2,745	95.6%
English	120	4.2%
Banglish	7	0.2%
<i>Age group</i>		
18–22 years	1,054	36.7%
23–27 years	696	24.2%
13–17 years	422	14.7%
28–34 years	418	14.6%
35–44 years	187	6.5%
45+ years	95	3.3%
<i>Gender</i>		
Male	2,063	71.8%
Female	716	24.9%
Other	93	3.2%

Several aspects of the dataset composition are noteworthy for interpreting the experimental results. First, the dataset is heavily skewed toward high-risk cases (46.7%), reflecting the clinical context of the application. Second, the dataset is predominantly young (60.9% aged 18–27), male (71.8%), and Bengali-speaking (95.6%), which limits generalisability of the empathy marker sets to other populations. Third, the high proportion of severe PHQ-4 scores (44.7% scoring 9–12) confirms that the dataset captures genuine clinical distress rather than mild wellbeing concerns.

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Institutional Review Board Statement: The dataset used in this study consists of de-identified prompts collected as part of routine mental-health screening and support services in Bangladesh. Users provided informed consent within the application at the point of data collection, acknowledging that anonymised interaction data may be used for research purposes. No personally identifiable information was retained. As the data were collected in the course of standard service delivery and fully anonymised prior to research use, formal ethical review was

not required under applicable institutional guidelines. The study was conducted in accordance with the Declaration of Helsinki.

Informed Consent Statement: Informed consent was obtained from all participants via the mental-health support application at the point of data collection. Users acknowledged that anonymised interaction data may be used for research purposes. No personally identifiable information was retained and no individual participants can be identified from the published data.

Data Availability Statement: The dataset used in this study is not publicly available due to patient privacy restrictions and the sensitive nature of mental-health screening data. The source code for the EvoRiri framework, including the genetic algorithm, response generator, and scoring functions, is available from the corresponding author upon reasonable request.

Conflicts of Interest: The corresponding author’s PhD supervisor, Prof. Mir Md. Jahangir Kabir, serves as Guest Editor of the Special Issue “Evolutionary Algorithms in Artificial Intelligence” in *Mathematics* to which this manuscript is submitted. This conflict of interest has been disclosed to the MDPI editorial office prior to submission in accordance with MDPI’s conflict of interest policy.

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